

Boundary Case: High-Frequency Mutation Environment — Governance Implications When LLM Version Changes Occur Very Frequently, Testing Verification Gate Execution Capacity, Action-Feedback Evidence Reliability, and High-Stakes Pinning Criticality at High Mutation Temporal Density

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize the architectural implications of deploying a CKS Self in a high-frequency mutation environment — where LLM version changes occur multiple times per day or several times per week — and to characterize where governance operational capacity approaches a structural limit under high mutation temporal density.

Abstract

The CKS architecture (Paper 2) specifies mutation governance instruments for every instinct evolution event. In typical deployments, LLM version changes are infrequent; mutation governance is an occasional activity. This note formalizes the boundary case in which LLM version changes occur at high frequency — multiple times per day or several times per week. At high mutation frequency, three architectural features are placed under pressure: (1) verification gate execution (B2.06) must run per mutation event, making it a continuous rather than episodic governance activity, with automation becoming necessary but remaining governed labor rather than autonomous governance; (2) action-feedback proposing substrates (B2.74) face an evidence window problem, where evidence accumulated per LLM version may be sparse while cross-version evidence windows conflate behavioral patterns from distinct versions, degrading evidence reliability; and (3) high-stakes pinning (B2.05) becomes increasingly critical as mutation frequency rises, because the value of insulating high-stakes decisions from instinct volatility scales directly with how rapidly instinct changes. The note identifies the architectural limit: the architecture specifies that governance must keep pace with mutation events; at frequencies where governance cannot keep pace, the deployment is operating outside its intended governance capacity and approaches the Ungoverned Mutation anti-pattern (B3.14).

1. Configuration Description

The deployment is a CKS Self whose instinct layer undergoes LLM version changes at high frequency — specifically, at a rate that may reach multiple changes per day or several changes per week. The mutation events are individually discrete: each constitutes one instinct evolution event, arriving from upstream (the

LLM provider or the substrate-platform infrastructure) rather than from goals the Self pursues. This undirected character is the defining property of instinct evolution per Paper 2 §7.2.

The deployment is not in violation of any architectural commitment. All mutation governance instruments per B1.13 are present and correctly applied per event. Verification substrates execute at each mutation boundary. Routing rules account for the current LLM version. High-stakes pinning is in effect for designated decision paths. The deployment's governance structure is correctly instantiated; the boundary under examination is not about correctness but about *operational capacity*: whether governance can keep pace with the frequency at which mutation events arrive.

The boundary is therefore a temporal density boundary. The architecture's mutation governance instruments were designed for mutation events, not for mutation events at arbitrarily high frequency. What this note formalizes is what happens to governance operational capacity as mutation frequency increases, and where the architecture reaches its intended limit.

2. Architectural Boundary Being Tested

Two architectural features are placed under simultaneous pressure at high mutation frequency.

B1.13 mutation governance at high frequency. Paper 2 §7.3 specifies multi-level simultaneous evolution, with mutation governance instruments applying to each instinct evolution event. The instruments include verification gate execution per B2.06, routing rule review per B2.04, and high-stakes pinning review per B2.05. Each instrument must execute per event. At low mutation frequency, these are episodic governance activities. At high mutation frequency — multiple times per day — each instrument must execute at corresponding frequency. The boundary being tested is whether governance execution capacity (human and automated together) can match mutation event density.

B1.15 action-feedback evidence reliability at high frequency. Paper 2 §7.2 specifies action-feedback evolution as the mechanism by which a Self learns from its own operation: lived experience in the action layer informs substrate refinement, feeding back to the DNA layer through proposal-and-acceptance machinery. The reliability of this mechanism depends on action evidence accurately representing the behavioral patterns of the current LLM version — which requires sufficient evidence to accumulate within a single version's tenure. At high mutation frequency, the time window during which any one LLM version is active may be shorter than the minimum window needed for reliable evidence accumulation. The boundary being tested is whether action-feedback evidence can remain reliable — accurately representing a single version's behavioral patterns — when versions change faster than evidence accumulates.

These two pressures interact. A deployment under high mutation frequency faces both simultaneously, and the governance configuration must address both.

3. Governance Implications

Four governance implications follow from high mutation frequency.

Verification gate automation necessity, as governed labor. At high mutation frequency, verification gates per B2.06 cannot be manually executed per event by human operators. If mutation events arrive multiple times per day, manual gate execution becomes the constraint that limits achievable mutation frequency. The

governance response is to automate verification gate execution: automated pipelines execute the gate criteria on each version transition, flag exceptions for human review, and record execution evidence per A2.40.

The critical framing is that this automation is *labor* in the sense of A1.12, not authority. Human governance configures the automated gates — specifying the criteria, the exception conditions, and the review thresholds — and humans review gate outputs at appropriate intervals. Automated execution performs the repetitive labor; it does not exercise governance authority. A deployment in which automated gates execute without human configuration or review would no longer be human-governed in the CKS sense: it would have delegated authority to automation rather than delegating labor. The governance implication is therefore not “automate and stand down” but “automate the labor while retaining the authority architecture that governs the automation.”

Routing strategy complexity under rapid version succession. Routing rules per B2.04 specify how work is directed based on LLM version characteristics. At high mutation frequency, the routing strategy must accommodate rapid version succession — configurations that were current at the start of the day may be superseded by nightfall. Version staging per B2.04 becomes more complex when multiple versions succeed each other within a single operational day. The governance implication is that routing substrates must be versioned, that routing rule updates must be executed as part of the per-mutation governance activity, and that routing substrate entries must carry the version-association metadata that makes routing decisions traceable to the version in effect at decision time.

Action-feedback evidence window configuration. Proposing substrates per B2.74 accumulate action-layer evidence and feed it forward to the DNA-layer proposal-and-acceptance machinery. The evidence window — the temporal scope over which evidence is accumulated — must be configured as an explicit governance decision at high mutation frequency.

A per-LLM-version evidence window preserves evidence reliability: evidence accumulated within a version window reflects only that version’s behavioral patterns, and the determinism commitment per A1.10 applies cleanly within the window. The cost is that each window may be short and may contain limited evidence, reducing the statistical confidence of action-feedback proposals.

A cross-version evidence window accumulates more evidence, raising statistical confidence, but at the cost of mixing behavioral patterns from multiple LLM versions. Mixed-version evidence degrades the B1.15 + A1.10 pair: the determinism contract applies within a version’s tenure, but cross-version evidence spans multiple distinct behavioral regimes. An action-feedback proposal developed from cross-version evidence may be responding to a version-specific pattern that no longer obtains, or to an averaged pattern that does not accurately describe any single version.

The governance implication is that evidence window configuration must be explicit rather than left to default, and the substrate must record which LLM version was in effect when each action-layer record was created. Proposing substrates must acknowledge which versions their evidence spans. Where cross-version evidence is used, the limitation must be recorded as part of the proposal record so that human reviewers can account for it in the proposal-and-acceptance process.

High-stakes pinning criticality scaling with mutation frequency. Paper 2 §8.3 specifies that high-stakes decisions can be architecturally pinned to the reasoning layer regardless of how capable instinct becomes.

The value of pinning is that pinned decisions are insulated from instinct volatility: they remain in the reasoning layer even when LLM version changes bring behavioral shifts in the instinct layer. At high mutation frequency, the instinct layer may be a different LLM version today than it was yesterday; behavioral patterns near high-stakes decisions shift accordingly. The insulation value of pinning scales directly with mutation frequency — a deployment undergoing multiple version changes per day benefits from pinning proportionally more than one undergoing one change per month. The governance implication is that high-stakes pinning review is among the most critical governance activities at high mutation frequency, and that comprehensive pinning coverage should be treated as a prerequisite for high-frequency mutation operation rather than an optional enhancement.

4. Boundary Tests

Three boundary tests determine whether a high-frequency mutation deployment is operating within intended governance capacity.

Boundary test (a): Verification gate execution density matching mutation event density. The verification gate execution record per A2.40 should contain one execution record per mutation event. At high mutation frequency, record density should match mutation event density — multiple records per day when multiple mutation events per day occur. A deployment in which mutation events outpace gate execution records is executing verification gates at a lower rate than the mutation rate, which is verification-skipped operation for the events without records. The test is: count mutation events in a given period; count verification gate execution records in the same period; the ratio should be 1:1. A ratio below 1:1 indicates governance capacity is already being exceeded.

Boundary test (b): Evidence window configuration appropriate to mutation frequency. Proposing substrates per B2.74 should carry explicit evidence window configuration. The test is: does the configuration specify whether windows are per-LLM-version or cross-version? If cross-version, does the configuration acknowledge the mixed-version limitation? Does each proposal record indicate which LLM versions contributed evidence to it? A deployment in which proposing substrates use cross-version evidence without recording version provenance is operating with unacknowledged evidence reliability limitations — the mixed-version risk exists but is invisible to governance reviewers.

Boundary test (c): High-stakes pinning coverage reviewed for the deployment's current mutation frequency. The pinning registry should enumerate the decision paths covered by B2.05. The test is: was pinning coverage configured at the current mutation frequency, or at a lower frequency that previously obtained? High-stakes designation is itself substrate content under governance; at high mutation frequency, the scope of what counts as high-stakes may need to expand relative to a prior lower-frequency configuration. A deployment whose pinning coverage was set when mutation frequency was low, and which has not been reviewed since frequency increased, may have underprotected decision paths — correct at the time of configuration, insufficient at the current mutation rate.

5. Stress Points

Two stress points identify where high-frequency mutation environments approach architectural failure modes.

Governance capacity overload and Ungoverned Mutation (B3.14). The primary stress point is where mutation frequency exceeds governance verification execution capacity. At that threshold, mutation events arrive faster than verification gates can be executed — not because the gates are absent, but because the cadence of execution cannot match the cadence of events. This produces Ungoverned Mutation Form 1 (verification-skipped) per B3.14: LLM versions become active without verification gate coverage for the specific transitions where execution lapses.

The stress point is not a binary threshold. It has a gradient. At modest mutation frequency, manual gate execution is possible. As frequency rises, manual execution becomes a bottleneck. Automation raises the capacity threshold but introduces its own limits: automated gates have a throughput ceiling, and configuration review by human operators introduces a separate constraint. The stress point is the combined rate at which total governance capacity — human and automated together — is saturated. Beyond that rate, governance cannot keep pace.

Characterizing the stress point for a given deployment requires mapping its governance capacity: how rapidly can automated gates execute? How frequently are human operators available to review gate outputs and exception flags? What is the minimum human review interval compatible with the authority retention that governed operation requires? These are deployment-specific parameters. The architecture specifies the governance structure; it does not bound the maximum supportable frequency. The note's contribution is to establish that such a capacity limit exists and to identify the failure mode — B3.14 Form 1 — when it is exceeded.

Evidence contamination at high mutation frequency. The secondary stress point is evidence contamination. As mutation frequency rises, the tenure of any single LLM version shrinks. Below a certain tenure duration, the action-feedback evidence accumulated within one version's window is insufficient for reliable proposals: the evidence is simply too sparse. If the governance response is to expand evidence windows across versions to compensate, the mixed-version contamination risk rises correspondingly.

Evidence contamination is most consequential for DNA-layer proposals, where action-feedback evolution feeds back to orchestration substrate updates. An orchestration substrate update based on contaminated evidence may correct for a version-specific behavioral pattern that no longer obtains, or may introduce a rule tailored to an averaged behavioral pattern that accurately describes no current version. Human reviewers operating on proposal records without version provenance metadata cannot detect this contamination. The governance mitigation — explicit evidence window configuration with version provenance recording — does not eliminate the underlying uncertainty, but makes it visible and accountable in the proposal record. Invisible contamination is the stress condition; visible, acknowledged contamination is the managed state that keeps action-feedback evolution operable at high mutation frequency.

6. Architectural Limits

The CKS architecture specifies mutation governance instruments per B1.13 and requires their application to each instinct evolution event. It does not specify a maximum mutation frequency. The architecture's specification implies that governance can keep pace with mutation events; it does not bound how rapidly governance can execute in any given deployment.

The practical frequency limit is determined by governance capacity — human and automated together. This is a deployment parameter, not an architectural one. Different deployments will have different limits depending on how automated verification gates are configured, how quickly human operators can review outputs, and what minimum review interval is compatible with the authority retention that human-governed operation requires.

The architectural limit this boundary case formalizes is not a hard frequency ceiling but a structural consequence: *at frequencies where governance cannot keep pace with mutation events, the architecture is being operated outside its intended governance capacity*. The mutation governance instruments specified by B1.13 presuppose their own executability per event; a deployment where execution cannot match frequency is not operating the instruments as specified, regardless of whether the instruments are formally present.

Two implications follow. First, high-frequency mutation deployments should characterize their governance capacity before operating at high frequency — identifying the throughput of automated gate execution, the availability of human reviewers, and the minimum review interval — rather than discovering the capacity limit empirically through governance lapses. Second, the architecture does not prohibit high-frequency mutation; it requires that governance scale correspondingly. If governance capacity can be expanded through automation, through streamlined review processes, or through more frequent human involvement, then the supportable frequency rises proportionally. The limit is governance capacity; the architecture’s requirement is that governance capacity match mutation frequency, not that frequency be held artificially low.

The high-frequency mutation boundary case is thus a design input for deployments planning to operate at high instinct evolution rates: before committing to a mutation cadence, characterize governance capacity, configure evidence windows explicitly for the expected version tenure, ensure pinning coverage is comprehensive at the planned frequency, and establish that automated verification gate execution can match the planned event density with human oversight retained over the automation.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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